Kangjoon Cho

Greater Boston, MA

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Applied Geospatial Scientist and Statistical Modeler with expertise in satellite time-series image analysis, remote sensing, statistical modeling, and machine learning for Earth systems and environmental applications. Experienced in developing end-to-end computational workflows that integrate high-dimensional data, uncertainty quantification, predictive modeling, and model validation. Strong background in geospatial analysis, terrestrial carbon modeling, life cycle assessment, and AI-enabled environmental and energy analysis, with applications to renewable solar energy development, land cover change, ecosystem monitoring, and sustainable infrastructure planning.

EDUCATION

Boston University, *College of Arts and Sciences* — **Boston, MA**

Doctor of Philosophy: Department of Earth and Environment, GPA 3.98/4.0 — May 2026
Advisor: Dr. Curtis E. Woodcock
Dissertation: Impact of Utility-Scale Solar Development on Land Cover and Carbon Dynamics in Massachusetts

Seoul National University, *College of Engineering* — **Seoul, South Korea**

Master of Science: Civil and Environmental Engineering, **Valedictorian**, GPA 4.0/4.0 — Aug. 2019
Advisor: Dr. Yongil Kim
Thesis: Modulation and Regression based Hybrid Thermal Sharpening of Landsat-8 TIRS Imagery Using Fractional Urban Cover

Seoul National University, *College of Engineering* — **Seoul, South Korea**

Bachelor of Science: Civil and Environmental Engineering, **Cum laude**, GPA 3.59/4.0 — Aug. 2017
Thesis: Precise Air Pollution Mapping using COMS Data

RESEARCH EXPERIENCE

Research Fellow — Boston, MA
Center for Remote Sensing, Boston University — Jan. 2021 – May 2026

- Conducted independent and collaborative research in satellite time-series analysis, remote sensing change detection, and environmental modeling across renewable-energy infrastructure, ecosystem disturbance, invasive species, and large-scale land-cover applications
- Built scalable cloud-based geospatial workflows to process and analyze 10,000+ Landsat and Sentinel images, integrating preprocessing, time-series feature extraction, machine learning classification, validation, and uncertainty analysis
- Integrated remote-sensing outputs with carbon, energy, and ecological models to translate observed land-cover change into decision-relevant estimates of environmental impact, net carbon benefit, and uncertainty

Research Assistant — Seoul, South Korea
Spatial Informatics & Systems Lab: Remote Sensing Group, Seoul National University — May 2016 – Aug. 2019

- Conducted multi-sensor remote sensing research across disaster response, agricultural monitoring, hyperspectral analysis, and satellite-based air pollution mapping
- Developed satellite image processing and machine learning workflows for classification, change detection, super-resolution, spectral analysis, and environmental variable estimation using optical, thermal infrared, SAR, hyperspectral, and geostationary satellite data
- Produced geospatial analysis outputs for funded research projects involving wildfire monitoring, tsunami damage assessment, crop classification, soil and vegetation condition analysis, and PM2.5 estimation

WORK EXPERIENCE

Research Associate — Seoul, South Korea
Engineering Research Institute, Seoul National University — Sep. 2019 – Aug. 2020

- Conducted government-commissioned geospatial research for national agencies, combining spatial information infrastructure planning, stakeholder coordination, and very-high-resolution satellite image analysis
- Supported policy-facing and national security applications of geospatial analysis by translating technical remote sensing and spatial data infrastructure findings into actionable planning materials and facility-level interpretation products

Food Service Specialist, *Republic of Korea Air Force*, Pyeongtaek, South Korea — Feb. 2012 –Feb. 2014

RESEARCH PROJECTS

Jan. 2023–Present: Research Fellow, **Scalable Geospatial Inventory and Net Carbon Modeling of Utility-Scale Solar Development and Land Cover Change**, Independent Dissertation Research

- Designed and led an independent doctoral research project to evaluate the land cover and net carbon impacts of utility-scale solar development in Massachusetts
- Developed Landsat time-series and machine learning to monitor decadal solar infrastructure construction and associated land cover change
- Quantified deforestation associated with solar development, including both within facility boundaries and adjacent forest clearing around solar sites

- Integrated satellite-derived land cover change with biomass data, forest growth, decay models, photovoltaic life cycle assessment, solar electricity generation modeling, and displaced carbon emissions
- Developed a dynamic carbon modeling to evaluate how installation year, technology performance, ecosystem carbon dynamics, and changing grid emissions affect net carbon benefits
- Applied Monte Carlo simulation and uncertainty propagation to estimate statewide carbon impacts, carbon payback time, and uncertainty bounds

May 2021–Dec. 2025: Research Fellow, **New Opportunities Using the Landsat Temporal Domain: Monitoring Ecosystem Health, Condition and Use**, USGS Landsat Science Team

- Analyzed the spread and management of buffelgrass, an invasive species that increases wildfire risk, in and around Saguaro National Park, Arizona, using satellite time-series analysis
- Linked temporal vegetation dynamics with management intervention records and applied causal inference to assess the effectiveness of buffelgrass control efforts
- Collected and interpreted reference data to evaluate Amazon forest disturbance using multi-sensor data fusion with Landsat and Sentinel observations

Nov. 2021–Dec. 2023: Research Fellow, **MUTATED – MODELING and UNDERSTANDING using TEMPORAL ANALYSIS of TRANSIENT EARTH DATA**, IARPA BAA:19-04

- Improved construction-site detection workflows by analyzing features from time-series analysis and identifying the spectral-temporal variables most sensitive to construction-related land cover change
- Evaluated change classification performance against validation datasets
- Supported multi-sensor analysis of large construction activities through the fusion of optical and radar remote sensing

Aug. 2019–Aug. 2020: Project Manager, **A Study on Establishment of Spatial Information Roadmap for Supporting the Infrastructure Construction of the Unified Korean Peninsula**, National Geographic Information Institute

- Served as the point of contact for a government-commissioned geospatial consulting project for the National Geographic Information Institute, coordinating stakeholder requirements, technical expectations, and deliverables for long-term spatial infrastructure planning under a future unified Korea scenario
- Conducted technical review and strategic assessment of geospatial infrastructure integration needs, including coordinate reference systems, geodetic control points, national base mapping, spatial data standards, data governance, and system interoperability between South and North Korea
- Developed phased recommendations for spatial data integration, map production timelines, institutional coordination, and long-term geospatial system architecture, translating technical findings into policy-relevant implementation strategies for a national government agency

Oct. 2019–Mar. 2020: Research Associate, **A Study on the Object Classification of Nuclear Facilities in Neighboring Countries Using Very-High Resolution Satellite Image and Object Based Image Analysis Software**, Korean Institute of Nuclear Nonproliferation and Control

- Developed object-based image analysis to detect and classify infrastructure features and activities at nuclear facility sites using very-high resolution satellite imagery, including WorldView-2, WorldView-3, and KOMPSAT

May 2019–Dec. 2019: Researcher, **Development and Demonstration of Next Generation Land Information Model Using Multi Sensor and GeoAI Technology**, Spatial Information Research Institution of Korea Land and Geospatial Informatix Corporation

- Applied object-based machine learning workflows to airborne and drone-based hyperspectral imagery to detect cadastral map changes and support GeoAI-based land cover and land use classification

Jun. 2016–May 2019: Researcher, **A Study on Hyperspectral Image Processing Techniques for Crop Type Classification and Quality Assessment**, National Research Foundation of Korea

- Proposed a new Hyperspectral Narrow Index to investigate soil moisture and organic matter content from ground-based hyperspectral data
- Simulated Sentinel-2 products using an airborne hyperspectral image and conducted an analysis of Top-of-Atmosphere and Bottom-of-Atmosphere reflectances to evaluate the effectiveness of the "Sen2cor" module for atmosphere correction on agricultural land

May 2017–Mar. 2018: Researcher, **Damage Assessment of Large-scale Complex Disaster using Satellite Constellation**, National Research Foundation of Korea

- Developed super-resolution for thermal image and land surface temperature mapping for wildfire condition monitoring
- Applied quad-polarimetric SAR analysis and object-based change detection to evaluate tsunami-related damage

May 2016–Oct. 2016: Primary Investigator, **Precise Air Pollution Mapping using COMS Data**, SNU Undergraduate Research Program from Research Affairs of Seoul National University

- Estimated large-scale PM2.5 using Aerosol Optical Depth (AOD) from COMS datasets and established the relationship between AOD image and PM2.5 reference data

LEADERSHIP EXPERIENCE

PhD Representative, *Graduate Student Association, Boston University, Boston, MA* Jan. 2022 – Aug. 2023

- Hosted weekly coffee hours to strengthen peer communication and represented the department in university-wide meetings

TEACHING EXPERIENCE

Teaching Fellow, *Boston University,* Boston, MA — Jan. 2024 – Dec. 2025

- Led weekly sessions for Introduction to Quantitative Environmental Modeling class
- Teach introductory R programming, multivariate regression, Bayesian stochastic modeling, and uncertainty quantification
- Mentored over 100 undergraduate students in applied quantitative modeling and data analysis

Peer Tutoring, *Seoul National University*, Seoul, South Korea — Mar. 2016 – Feb. 2017

- Tutored peers in the Department of Civil and Environmental Engineering supported by the College peer tutor program
- Taught Calculus, Statistics, Physics, Mechanics of Materials

HONORS AND AWARDS

Commendation Letter

- Korean Society of Surveying, Geodesy, Photogrammetry and Cartography

Best Paper Award

- Grand Prize, The Paper Contest of Writing in Science & Technology, Faculty of Liberal Education, Seoul National University

TECHNICAL SKILLS

Programming: R, Python, JavaScript, MATLAB, C++, SQL

Geospatial & Remote Sensing: Google Earth Engine, GIS, Geospatial time-series analysis, Image segmentation, Data clustering, Super-resolution, Digital image processing; Optical, infrared, thermal, and radar image processing

Modeling & Statistics: Machine learning, Monte Carlo simulation, uncertainty propagation, Signal processing, Multivariate & Non-linear Regression, Bayesian inference, MCMC, Predictive modeling

Energy & Carbon Modeling: Statewide solar electricity modeling (PVWatts/System Advisor Model), Life cycle assessment, Avoided emissions, Terrestrial carbon modeling, Carbon bookkeeping

Computing: Cloud-based geospatial processing (Google Earth Engine, AWS, research cloud), Git/GitHub

PEER-REVIEWED PUBLICATIONS

Gu, H., Tang, X., **Cho, K.**, Acord, A. J., Rasmussen, P. G., Bosch, M. & Woodcock, C. E. Detection of New Construction Activities with Sentinel-1 and Landsat Time Series. *Remote Sensing Applications: Society and Environment*. 2026, 41, p.101893.

Cho, K. & Woodcock, C. E. Detecting Utility-Scale Solar Installations and Associated Land Cover Changes Using Spatiotemporal Segmentation of Landsat Imagery. *Science of Remote Sensing*. 2025, p.100337.

Tang, X., Barrett, M.G., **Cho, K.**, Bratley, K.H., Tarrio, K., Zhang, Y., Gu, H., Rasmussen, P., Bosch, M. & Woodcock, C.E., Broad-area-search of new construction using time series analysis of Landsat and Sentinel-2 data. *Science of Remote Sensing*, 2024, p.100138.

Tang, X., Bratley, KH., **Cho, K.**, Bullock, E., Olofsson, P. & Woodcock, CE. Near-real-time monitoring of tropical forest disturbance by fusion of Landsat, Sentinel-2, and Sentinel-1 data. *Remote Sensing of Environment*. 2023, 294, 113626.

Kim, M., **Cho, K.**, Kim, H. & Kim, Y. Spatiotemporal fusion of high resolution land surface temperature using thermal sharpened images from regression-based urban indices. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*. 2020, 3, 241-254.

Cho, K. & Kim, Y. Simulation of Sentinel-2 Product Using Airborne Hyperspectral Image and Analysis of TOA and BOA Reflectance for Evaluation of Sen2cor Atmosphere Correction: Focused on Agricultural Land. *Korean Journal of Remote Sensing*. 2019, 35(2), 251-263. (In Korean with English abstract)

Cho, K., Kim, Y. & Kim, Y. Disaggregation of Landsat-8 Thermal Data Using Guided SWIR Imagery on the Scene of a Wildfire. *Remote Sensing*. 2018, 10, 105.

MANUSCRIPTS IN PREPARATION

Cho, K., Stoynova, H., Thompson, J., Cleveland, C., Hutyra, L., Tang, X. & Woodcock, C. E. Impact of Utility-Scale Solar Development on Land Use and Carbon Dynamics in Massachusetts. *Global Environmental Change.*

CONFERENCE AND SEMINAR

Bratley, K. H., Nolte, C., Kaufmann, R., **Cho, K.** & Woodcock, C. E. (2024, December). Causal Inference in Ecological Management: Evaluating Treatment Efficacy on Invasive Buffelgrass (Cenchrus ciliaris) in Saguaro National Park Using Landsat Time-Series and Panel Regression Modeling. In AGU Fall Meeting Abstracts. (Vol. 2024, No. 1517, pp. B11N-1517)

Cho, K. & Woodcock, C. E. Mapping Land Cover Changes Caused by Utility-Scale Solar Installation using the Continuous Change Detection and Classification - Spectral Mixture Analysis (CCDC-SMA) Algorithm. AGU Fall Meeting 2023, San Francisco, United States, 11-15 December 2023.

Tang, X., Gu, H., **Cho, K.**, Barrett, M., Acord, A., Rasmussen, P., Bosch, M. & Woodcock, C. E. Broad-area-search of new constructions using time series analysis of Landsat and Sentinel-1 data. AGU Fall Meeting 2023, San Francisco, United States, 11-15 December 2023.

Tang, X., Bratley, KH., **Cho, K.**, Bullock, E., Olofsson, P. & Woodcock, C. E. Near-real-time monitoring of tropical forest disturbance by fusion of Landsat, Sentinel-1, and Sentinel-2 data. AGU Fall Meeting 2021, New Orleans, United States, 13-17 December 2021.

Cho, K., Kim, Y. & Kwak, T. Object Based Hyperspectral Image Analysis for Cadastral Mapping. The 40th Asian Conference on Remote Sensing, Daejeon, South Korea, 14-18 October 2019.

Cho, K. & Kim, Y. Sharpening Algorithm of Landsat-8 TIR Data for the Urban Area. International Symposium on Remote Sensing 2019, Taipei, Taiwan, 17-19 April 2019.

Kim, Y., **Cho, K.**, Oh, J. & Kim, Y. The 2011 Tohoku Earthquake and Tsunami-induced Coastal Damage Assessment using L-band Polarimetric Features using Random Forest Classifier: An efficient-oriented reduction strategy. The 3rd International Water Safety Symposium, Incheon, South Korea, 19-23 June 2018.

Cho, K. & Kim, Y. Estimation of Land Surface Temperature for Assessing Urban Heat Island Effect. International Symposium on Remote Sensing 2018, Pyeongchang, South Korea, 09-11 May 2018.

Cho, K., Kim, Y. & Kim, Y. Quantitative Comparison of Sentinel-2 Level-2A Product and Simulated Imagery from Airborne Hyperspectral Data. ESA 2nd Sentinel-2 Validation Team Meeting, Rome, Italy, 29-31 January 2018.

Cho, K., Kim, Y. & Kim, Y. Disaster Damage Estimation using Change Detection Method with PolSAR Imagery and Object based Analysis. The 2017 Fall Conference of the Korea Society of Remote Sensing, Yesan, South Korea, 02-03 November 2017. (In Korean)

Cho, K., Kim, Y. & Kim, Y. Soil Condition Analysis using a Ground-based Hyperspectral Data. International Symposium on Remote Sensing 2017, Nagoya, Japan, 17-19 May 2017.

Cho, K., Kim, Y. & Kim, Y. Mapping of Fine Particulate Matter Concentration Using COMS Data. The 2016 Fall Conference of the Korea Society of Remote Sensing, Chungju, South Korea, 03-04 November 2016. (In Korean)

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